The Journal of the Acoustical Society of America

How Understanding Voice Naturalness Includes Authenticity and Further Aspects of Self-Voice Processing: Comment on "Beyond acoustics: Self-relevance as a key to voice naturalness", J. Acoust. Soc. Am. 158, 4045–4047 (2025).

--Manuscript Draft--

Manuscript Number:	JASA-13582R1
Full Title:	How Understanding Voice Naturalness Includes Authenticity and Further Aspects of Self-Voice Processing: Comment on "Beyond acoustics: Self-relevance as a key to voice naturalness", J. Acoust. Soc. Am. 158, 4045–4047 (2025).
Article Type:	Comment
Corresponding Author:	Christine Nussbaum Friedrich Schiller University Jena: Friedrich-Schiller-Universitat Jena GERMANY
First Author:	Christine Nussbaum
Order of Authors:	Christine Nussbaum
	Sascha Frühholz
	Stefan R. Schweinberger
Keywords:	voice; naturalness; self-voice; authenticity
Additional Information:	
Question	**Response**
Please read the Transfer of Copyright Agreement and the Instructions. By selecting the appropriate response below, you verify that you have consented to the Transfer of Copyright Agreement (if applicable) and the conditions and representations set forth in the Instructions.	R - Regular
Please provide the DOI link to the published JASA article being commented on.	https://doi.org/10.1121/10.0039927

How *Understanding Voice Naturalness* Includes Authenticity and Further Aspects of Self-Voice Processing: Comment on "Beyond acoustics: Self-relevance as a key to voice naturalness", J. Acoust. Soc. Am. 158, 4045–4047 (2025).

Christine Nussbaum[1,2], Sascha Frühholz[3,4], and Stefan R. Schweinberger[1,2,5]

[1]Department for General Psychology and Cognitive Neuroscience, Friedrich Schiller University Jena, 07743 Jena, Germany

[2]Voice Research Unit, Friedrich Schiller University, 07743 Jena, Germany

[3]Department of Psychology, University of Oslo, 0371 Oslo, Norway

[4]Cognitive and Affective Neuroscience Unit, University of Zurich, 8050 Zurich, Switzerland

[5]Swiss Center for Affective Sciences, University of Geneva, 1222 Geneva, Switzerland

Correspondence should be addressed to Christine Nussbaum (https://www.fsv.uni-jena.de/en/63723/christine-nussbaumatuni-jena-de), Department for General Psychology and Cognitive Neuroscience, Friedrich Schiller University Jena, Am Steiger 3/1, 07743 Jena, Germany. Tel: +49 (0) 3641 945934, E-Mail: christine.nussbaum@uni-jena.de.

Abstract: Understanding voice naturalness is of substantial practical importance to very different applied fields, including voice manipulation, synthesis, and pathology. We proposed an inclusive framework to unify manifold research efforts under a concise conceptualization of voice naturalness (Nussbaum et al., 2025). A recent letter by Pinheiro (2025) criticized that our framework (1) overlooks the self-voice and (2) reduces naturalness to considerations of acoustics alone. In this comment, we unpack these criticisms, and we resolve these potential misunderstandings by showing how self-voice research is easily integrated into our framework and especially its concept of authenticity.

What makes a voice sound *natural*? This is a question of high relevance across many domains: in speech-language pathology, across the manifold possibilities of voice modification, and in voice synthesis and cloning, a rapidly developing field that is now invading everyday life from various directions (Lavan et al., 2025). However, a related and equally important question received relatively little attention until now (Nussbaum et al., 2025; Pandey et al., 2025): What do we *mean* by voice naturalness? To address this conceptual gap, we recently published a framework for voice naturalness, which offers a concise definition that is rooted in theoretical models on voice perception and is flexibly applicable to diverse empirical contexts (Nussbaum et al., 2025). In a nutshell, we proposed a taxonomy with two distinct types: deviation-based naturalness and human-likeness-based naturalness. The deviation-based approach assesses naturalness in terms of distance away from a reference, while the human-likeness-based approach assesses naturalness according to its similarity to a specific reference – the human voice. With voice naturalness being a diverse and highly interdisciplinary topic, we are pleased to see that our work has already inspired a deeper discussion on its widespread implications.

In a recent letter to the editor, Pinheiro (2025) highlighted the importance of naturalness regarding the perception of one's own voice. A key point in this letter is that impressions of one's own voice (including synthesized versions of it) as sounding "natural to me" are the result of a complex cognitive process that cannot be explained by the acoustics of the voice signal alone. Instead, impressions of self-voice are based on self-relevance, defined by Pinheiro as the alignment between that voice and the listener's internal self-representation. Pinheiro (2025) claims that this is an aspect that our framework "overlooked", and the author proposes self-relevance as a complementary dimension for voice naturalness.

We welcome the opportunity to comment on Pinheiro´s letter. Of course, there is no doubt that the discussion around the perception of one´s own voice touches important issues in voice research, and we agree that impressions of naturalness in self-voice evaluation deserve more attention. Above all, however, we want to emphasize that this is not in conflict but in complete accordance with our proposed framework. The points raised by Pinheiro (2025) neither reveal a blind spot nor an incomplete conceptualization of voice naturalness in Nussbaum et al. (2025). In the next two paragraphs, we resolve these potential misunderstandings and show how self-voice research is easily integrated into our framework and especially its concept of authenticity. In the final paragraph, we briefly expand on the important points Pinheiro raised around current challenges around research on self-voice processing.

Voice naturalness – a matter of subjective impressions

Defining naturalness is challenging, and was a key objective in Nussbaum et al. (2025). Pinheiro (2025) wrote that our proposed definition of "deviation-based naturalness captures how much a voice diverges from typical acoustic patterns" (p. 4045). This is not fully correct. In fact, we refer to the "deviation from a reference that represents maximum naturalness" (Nussbaum et al., 2025, p. 472). Although the difference may seem small, it is crucial because it captures the key point raised by Pinheiro (2025): that impressions of naturalness are influenced by many aspects that go beyond acoustics alone. As we outlined, it is common for listeners to use an internal implicit reference based on their experience and expectations (Nussbaum et al., 2025, p. 472), and is therefore formed in "the ear of the beholder" (for empirical evidence, see e.g. Kapolowicz et al., 2022; Lee, 2010). Any paraphrasing of voice naturalness that omits this subjective component is not only scientifically incomplete, but potentially harmful, because the implication – even if unintentional - that there is an objective ground truth of what constitutes naturalness in voices gives rise to discrimination and repression of vocal diversity. Vocal minorities, who locate themselves outside of the distribution of "typical" acoustic patterns, would be labelled as unnatural, although they may simply express their natural individuality. To counteract such potential harm, we share the responsibility in making an active effort though our scientific debates by reflecting on the choice of wording and potential misunderstandings that could arise from it. Note that for the same reason, we also recommend caution with words like "appropriateness" as a reconceptualization of naturalness (Pinheiro, 2025, p. 4046), which may carry an additional evaluative layer which could be perceived as hurtful by the affected individuals, despite best intentions. On the conceptual understanding beneath, however, Pinheiro (2025) sums it up perfectly: “Naturalness […] must therefore be understood as emerging from the interaction between the acoustic properties of speech and the listener’s representational framework.” (page 4046). We are fully aligned with this - in fact, it is the essence of our model (cf. Figure 2, Nussbaum et al. 2025, page 473). We maintain that a full understanding of voice naturalness requires more systematic research that assesses differences across listeners and contexts, addressing societal aspects of speech production as well.

Linking voice naturalness and authenticity to self-voice processing

Another point raised by Pinheiro (2025) was that Nussbaum et al. (2025) “overlooked the self-voice”. Note that we intentionally decided to only briefly touch on this issue, but it provided substantial motivation for understanding voice naturalness: we discussed the seminal work by Yamagishi et al. (2012) and Hyppa-Martin et al. (2024), who evaluate personalized speech-synthesis technology for individuals who lost their biological voice (e.g., due to laryngectomy) and outlined how important it is that these devices sound authentic to both the speaker and their next of kin, like “their own natural voice”.

Nevertheless, we agree with Pinheiro’s call for more scientific attention towards multifaceted aspects that contribute to the impression of a voice as one’s own. Closer contact with Nussbaum et al. (2025) reveals how this call can be easily integrated into the existing framework. Specifically, our distinction between *naturalness* and *authenticity* represents an obvious entry point (cf. Figure 3, Nussbaum et al. 2025, page 475). Whereas naturalness refers to the holistic impression of a voice (i.e. “does a voice sound natural or unnatural?”), authenticity has been used to refer to specific social signals, such as emotion, gender or identity (i.e. “does this voice sound like an authentic or posed expression of anger?”). Identity authenticity, specifically, is assessed with regard to a specific speaker: “does this voice sound authentically like speaker X?” (e.g.,Roswandowitz et al., 2024). Self-voice assessments can be seen as a special case of identity authenticity: “Does this voice authentically sound *like me*?” Note that this does not imply that self-voice assessment is equivalent to assessment of authenticity for other speakers' identities; their commonalities and differences require further empirical investigation. Pinheiro (2025) listed several reasons why the self-voice could be a special case (see also Frühholz & Schweinberger, 2021). To this end, the consistent conceptual integration of self-voice into the framework of naturalness, and authenticity in particular, offers a valuable anchor point for empirical designs that target aspects of impressions that are unique to the self-voice.

Current Challenges for Understanding Self-Voice Processing

In the last section, we briefly complement the points raised by Pinheiro (2025) by reflecting on current challenges for self-voice research, which are relevant for voice naturalness and beyond. In fact, there are intriguing parallels to voice naturalness research: while highly relevant, previous research on the self-voice has been fragmentary and unsystematic, and hampered by methodological and conceptual challenges. For instance, recordings of the own voice notoriously sound unnatural for a listener, because they lack bone conduction cues (Maurer & Landis, 1990). To become relevant for naturalness research, this unnaturalness of the own voice needs to be considered by presenting stimuli with special devices (i.e., bone-conducting headphones). The vast majority of published papers on the self-voice (including a recent one by Pinheiro et al., 2023) does not consider this issue in their experimental setup, although a few notable exceptions now exist (e.g. Orepic et al., 2023).

Another topic that is related to the self-voice, not mentioned by Pinheiro (2025), is kinship recognition, which is deeply rooted in our biology (Mateo, 2015) and well-studied in the face domain (Maloney & Dal Martello, 2006), but virtually unstudied in human voice perception. There are very good reasons to believe that perceived self-similarity is one of the key mechanisms of kinship recognition (DeBruine et al., 2009; Tsuchiya & Schweinberger, 2022), and that brain mechanisms for discriminating self and kin from unfamiliar persons may be partially overlapping (Platek & Kemp, 2009).

Finally, even after decades of self-*face* perception research, the question of whether the self is special does remain controversial in psychology, neuroscience and philosophy (Kovács et al., 2025; Sui & Humphreys, 2017). Against the background of far fewer empirical studies, proposals that self-*voice* perception is special are intriguing, but require more systematic evidence, as well as an integrative framework. We explicitly acknowledge recent work in that direction (Orepic & Pinheiro, 2026), although it is regrettable that it does not make links to the conceptual framework for voice naturalness. We add that future consideration of kinship perception might promote further understanding of a putative relationship between naturalness and self-relevance. We look forward to empirical contributions that increase our understanding of both self-voice processing and voice naturalness impressions.

Acknowledgments

We thank Manuel Pöhlmann and three anonymous reviewers for helpful suggestions on a previous version of this work. The authors gratefully acknowledge the award of funding through an EU-MSCA doctoral network "Voice Communication Sciences" (action 101168998).

Author Declarations

The authors have no conflicts to disclose.

Data Availability

There is no data linked to this manuscript.

References

DeBruine, L. M., Smith, F. G., Jones, B. C., Roberts, S. C., Petrie, M., & Spector, T. D. (2009). Kin recognition signals in adult faces. *Vision Res*, *49*(1), 38–43. https://doi.org/10.1016/j.visres.2008.09.025

Frühholz, S., & Schweinberger, S. R. (2021). Nonverbal auditory communication - Evidence for integrated neural systems for voice signal production and perception. *Progress in Neurobiology*, *199*, 101948. https://doi.org/10.1016/j.pneurobio.2020.101948

Hyppa-Martin, J., Lilley, J., Chen, M., Friese, J., Schmidt, C., & Bunnell, H. T. (2024). A large-scale comparison of two voice synthesis techniques on intelligibility, naturalness, preferences, and attitudes toward voices banked by individuals with amyotrophic lateral sclerosis. *Augmentative and Alternative Communication*, *40*(1), 31–45. https://doi.org/10.1080/07434618.2023.2262032

Kapolowicz, M. R., Guest, D. R., Montazeri, V., Baese-Berk, M. M., & Assmann, P. F. (2022). Effects of Spectral Envelope and Fundamental Frequency Shifts on the Perception of Foreign-Accented

Speech. *Language and Speech*, *65*(2), 418–443. https://doi.org/10.1177/00238309211029679

Kovács, G., Göschel, L., Pawlik, S. M., & Tramacere, A. (2025). The neural dynamics of current and past self-face perception: Challenging the privilege access hypothesis. *Cortex; a Journal Devoted to the Study of the Nervous System and Behavior*, *194*, 77–90. https://doi.org/10.1016/j.cortex.2025.11.006

Lavan, N., Irvine, M., Rosi, V., & McGettigan, C. (2025). Voice clones sound realistic but not (yet) hyperrealistic. *PLoS One*, *20*(9), e0332692. https://doi.org/10.1371/journal.pone.0332692

Lee, E.-J. (2010). The more humanlike, the better? How speech type and users' cognitive style affect social responses to computers. *Computers in Human Behavior*, *26*(4), 665–672. https://doi.org/10.1016/j.chb.2010.01.003

Maloney, L. T., & Dal Martello, M. F. (2006). Kin recognition and the perceived facial similarity of children. *Journal of Vision*, *6*(10), 1047–1056. https://doi.org/10.1167/6.10.4

Mateo, J. M. (2015). Perspectives: Hamilton's Legacy: Mechanisms of Kin Recognition in Humans. *Ethology*, *121*(5), 419–427. https://doi.org/10.1111/eth.12358

Maurer, D., & Landis, T. (1990). Role of bone conduction in the self-perception of speech. *Folia Phoniatrica*, *42*(5), 226–229. https://doi.org/10.1159/000266070

Nussbaum, C., Frühholz, S., & Schweinberger, S. R. (2025). Understanding voice naturalness. *Trends in Cognitive Sciences*, *29*(5), 467–480. https://doi.org/10.1016/j.tics.2025.01.010

Orepic, P., Kannape, O. A., Faivre, N., & Blanke, O. (2023). Bone conduction facilitates self-other voice discrimination. *Royal Society Open Science*, *10*(2), 221561. https://doi.org/10.1098/rsos.221561

Orepic, P., & Pinheiro, A. P. (2026). From Voice to Self: An Integrative Framework on Self-Voice Processing. *Perspect Psychol Sci*, 17456916261422585. https://doi.org/10.1177/17456916261422585

Pandey, A., Le Maguer, S., & Harte, N. (2025). What is Naturalness? In *13th edition of the Speech Synthesis Workshop* (pp. 215–221). ISCA. https://doi.org/10.21437/SSW.2025-33

Pinheiro, A. P. (2025). Beyond acoustics: Self-relevance as a key to voice naturalness (L). *The Journal of the Acoustical Society of America*, *158*(5), 4045–4047. https://doi.org/10.1121/10.0039927

Pinheiro, A. P., Sarzedas, J., Roberto, M. S., & Kotz, S. A. (2023). Attention and emotion shape self-voice prioritization in speech processing. *Cortex; a Journal Devoted to the Study of the Nervous System and Behavior*, *158*, 83–95. https://doi.org/10.1016/j.cortex.2022.10.006

Platek, S. M., & Kemp, S. M. (2009). Is family special to the brain? An event-related fMRI study of familiar, familial, and self-face recognition. *Neuropsychologia*, *47*(3), 849–858. https://doi.org/10.1016/j.neuropsychologia.2008.12.027

Roswandowitz, C., Kathiresan, T., Pellegrino, E., Dellwo, V., & Frühholz, S. (2024). Cortical-striatal brain network distinguishes deepfake from real speaker identity. *Communications Biology*, *7*(1), 711. https://doi.org/10.1038/s42003-024-06372-6

Sui, J., & Humphreys, G. W. (2017). The ubiquitous self: What the properties of self-bias tell us about the self. *Ann N Y Acad Sci*, *1396*(1), 222–235. https://doi.org/10.1111/nyas.13197

Tsuchiya, A., & Schweinberger, S. R. (2022). Erkennen von Verwandtschaft zu sich Selbst und bei Anderen [Recognition of Kinship to Self and in Others]: Part 2, Issue 1/2022. *In-Mind Magazine*, *1*(2). https://de.in-mind.org/article/erkennen-von-verwandtschaft-zu-sich-selbst-und-bei-anderen

Yamagishi, J., Veaux, C., King, S., & Renals, S. (2012). Speech synthesis technologies for individuals with vocal disabilities: Voice banking and reconstruction. *Acoustical Science and Technology*, *33*(1), 1–5. https://doi.org/10.1250/ast.33.1

How *Understanding Voice Naturalness* Includes Authenticity and Further Aspects of Self-Voice Processing: Comment on "Beyond acoustics: Self-relevance as a key to voice naturalness", J. Acoust. Soc. Am. 158, 4045–4047 (2025).

Christine Nussbaum[1,2], Sascha Frühholz[3,4], and Stefan R. Schweinberger[1,2,5]

[1]Department for General Psychology and Cognitive Neuroscience, Friedrich Schiller University Jena, 07743 Jena, Germany

[2]Voice Research Unit, Friedrich Schiller University, 07743 Jena, Germany

[3]Department of Psychology, University of Oslo, 0371 Oslo, Norway

[4]Cognitive and Affective Neuroscience Unit, University of Zurich, 8050 Zurich, Switzerland

[5]Swiss Center for Affective Sciences, University of Geneva, 1222 Geneva, Switzerland

Correspondence should be addressed to Christine Nussbaum (https://www.fsv.uni-jena.de/en/63723/christine-nussbaumatuni-jena-de), Department for General Psychology and Cognitive Neuroscience, Friedrich Schiller University Jena, Am Steiger 3/1, 07743 Jena, Germany. Tel: +49 (0) 3641 945934, E-Mail: christine.nussbaum@uni-jena.de.

Abstract: Understanding voice naturalness is of substantial practical importance to very different applied fields, including voice manipulation, synthesis, and pathology. We proposed an inclusive framework to unify manifold research efforts under a concise conceptualization of voice naturalness (Nussbaum et al., 2025). A recent letter by Pinheiro (2025) criticized that our framework (1) overlooks the self-voice and (2) reduces naturalness to considerations of acoustics alone. In this comment, we unpack these criticisms, and we resolve these potential misunderstandings by showing how self-voice research is easily integrated into our framework and especially its concept of authenticity.

What makes a voice sound *natural*? This is a question of high relevance across many domains: in speech-language pathology, across the manifold possibilities of voice modification, and in voice synthesis and cloning, a rapidly developing field that is now invading everyday life from various directions (Lavan et al., 2025). However, a related and equally important question received relatively little attention until now (Nussbaum et al., 2025; Pandey et al., 2025): What do we *mean* by voice naturalness? To address this conceptual gap, we recently published a framework for voice naturalness, which offers a concise definition that is rooted in theoretical models on voice perception and is flexibly applicable to diverse empirical contexts (Nussbaum et al., 2025). In a nutshell, we proposed a taxonomy with two distinct types: deviation-based naturalness and human-likeness-based naturalness. The deviation-based approach assesses naturalness in terms of distance away from a reference, while the human-likeness-based approach assesses naturalness according to its similarity to a specific reference – the human voice. With voice naturalness being a diverse and highly interdisciplinary topic, we are pleased to see that our work has already inspired a deeper discussion on its widespread implications.

In a recent letter to the editor, Pinheiro (2025) highlighted the importance of naturalness regarding the perception of one's own voice. A key point in this letter is that impressions of one's own voice (including synthesized versions of it) as sounding "natural to me" are the result of a complex cognitive process that cannot be explained by the acoustics of the voice signal alone. Instead, impressions of self-voice are based on self-relevance, defined by Pinheiro as the alignment between that voice and the listener's internal self-representation. Pinheiro (2025) claims that this is an aspect that our framework "overlooked", and the author proposes self-relevance as a complementary dimension for voice naturalness.

We welcome the opportunity to comment on Pinheiro´s letter. Of course, there is no doubt that the discussion around the perception of one´s own voice touches important issues in voice research, and we agree that impressions of naturalness in self-voice evaluation deserve more attention. Above all, however, we want to emphasize that this is not in conflict but in complete accordance with our proposed framework. The points raised by Pinheiro (2025) neither reveal a blind spot nor an incomplete conceptualization of voice naturalness in Nussbaum et al. (2025). In the next two paragraphs, we resolve these potential misunderstandings and show how self-voice research is easily integrated into our framework and especially its concept of authenticity. In the final paragraph, we briefly expand on the important points Pinheiro raised around current challenges around research on self-voice processing.

Voice naturalness – a matter of subjective impressions

Defining naturalness is challenging, and was a key objective in Nussbaum et al. (2025). Pinheiro (2025) wrote that our proposed definition of "deviation-based naturalness captures how much a voice diverges from typical acoustic patterns" (p. 4045). This is not fully correct. In fact, we refer to the "deviation from a reference that represents maximum naturalness" (Nussbaum et al., 2025, p. 472). Although the difference may seem small, it is crucial because it captures the key point raised by Pinheiro (2025): that impressions of naturalness are influenced by many aspects that go beyond acoustics alone. As we outlined, it is common for listeners to use an internal implicit reference based on their experience and expectations (Nussbaum et al., 2025, p. 472), and is therefore formed in "the ear of the beholder" (for empirical evidence, see e.g. Kapolowicz et al., 2022; Lee, 2010). Any paraphrasing of voice naturalness that omits this subjective component is not only scientifically incomplete, but potentially harmful, because the implication – even if unintentional - that there is an objective ground truth of what constitutes naturalness in voices gives rise to discrimination and repression of vocal diversity. Vocal minorities, who locate themselves outside of the distribution of "typical" acoustic patterns, would be labelled as unnatural, although they may simply express their natural individuality. To counteract such potential harm, we share the responsibility in making an active effort though our scientific debates by reflecting on the choice of wording and potential misunderstandings that could arise from it. Note that for the same reason, we also recommend caution with words like "appropriateness" as a reconceptualization of naturalness (Pinheiro, 2025, p. 4046), which may carry an additional evaluative layer which could be perceived as hurtful by the affected individuals, despite best intentions. On the conceptual understanding beneath, however, Pinheiro (2025) sums it up perfectly: “Naturalness […] must therefore be understood as emerging from the interaction between the acoustic properties of speech and the listener’s representational framework.” (page 4046). We are fully aligned with this - in fact, it is the essence of our model (cf. Figure 2, Nussbaum et al. 2025, page 473). We maintain that a full understanding of voice naturalness requires more systematic research that assesses differences across listeners and contexts, addressing societal aspects of speech production as well.

Linking voice naturalness and authenticity to self-voice processing

Another point raised by Pinheiro (2025) was that Nussbaum et al. (2025) “overlooked the self-voice”. Note that we intentionally decided to only briefly touch on this issue, but it provided substantial motivation for understanding voice naturalness: we discussed the seminal work by Yamagishi et al. (2012) and Hyppa-Martin et al. (2024), who evaluate personalized speech-synthesis technology for individuals who lost their biological voice (e.g., due to laryngectomy) and outlined how important it is that these devices sound authentic to both the speaker and their next of kin, like “their own natural voice”.

Nevertheless, we agree with Pinheiro’s call for more scientific attention towards multifaceted aspects that contribute to the impression of a voice as one’s own. Closer contact with Nussbaum et al. (2025) reveals how this call can be easily integrated into the existing framework. Specifically, our distinction between *naturalness* and *authenticity* represents an obvious entry point (cf. Figure 3, Nussbaum et al. 2025, page 475). Whereas naturalness refers to the holistic impression of a voice (i.e. “does a voice sound natural or unnatural?”), authenticity has been used to refer to specific social signals, such as emotion, gender or identity (i.e. “does this voice sound like an authentic or posed expression of anger?”). Identity authenticity, specifically, is assessed with regard to a specific speaker: “does this voice sound authentically like speaker X?” (e.g.,Roswandowitz et al., 2024). Self-voice assessments can be seen as a special case of identity authenticity: “Does this voice authentically sound *like me*?” Note that this does not imply that self-voice assessment is equivalent to assessment of authenticity for other speakers' identities; their commonalities and differences require further empirical investigation. Pinheiro (2025) listed several reasons why the self-voice could be a special case (see also Frühholz & Schweinberger, 2021). To this end, the consistent conceptual integration of self-voice into the framework of naturalness, and authenticity in particular, offers a valuable anchor point for empirical designs that target aspects of impressions that are unique to the self-voice.

Current Challenges for Understanding Self-Voice Processing

In the last section, we briefly complement the points raised by Pinheiro (2025) by reflecting on current challenges for self-voice research, which are relevant for voice naturalness and beyond. In fact, there are intriguing parallels to voice naturalness research: while highly relevant, previous research on the self-voice has been fragmentary and unsystematic, and hampered by methodological and conceptual challenges. For instance, recordings of the own voice notoriously sound unnatural for a listener, because they lack bone conduction cues (Maurer & Landis, 1990). To become relevant for naturalness research, this unnaturalness of the own voice needs to be considered by presenting stimuli with special devices (i.e., bone-conducting headphones). The vast majority of published papers on the self-voice (including a recent one by Pinheiro et al., 2023) does not consider this issue in their experimental setup, although a few notable exceptions now exist (e.g. Orepic et al., 2023).

Another topic that is related to the self-voice, not mentioned by Pinheiro (2025), is kinship recognition, which is deeply rooted in our biology (Mateo, 2015) and well-studied in the face domain (Maloney & Dal Martello, 2006), but virtually unstudied in human voice perception. There are very good reasons to believe that perceived self-similarity is one of the key mechanisms of kinship recognition (DeBruine et al., 2009; Tsuchiya & Schweinberger, 2022), and that brain mechanisms for discriminating self and kin from unfamiliar persons may be partially overlapping (Platek & Kemp, 2009).

Finally, even after decades of self-*face* perception research, the question of whether the self is special does remain controversial in psychology, neuroscience and philosophy (Kovács et al., 2025; Sui & Humphreys, 2017). Against the background of far fewer empirical studies, proposals that self-*voice* perception is special are intriguing, but require more systematic evidence, as well as an integrative framework. We explicitly acknowledge recent work in that direction (Orepic & Pinheiro, 2026), although it is regrettable that it does not make links to the conceptual framework for voice naturalness. We add that future consideration of kinship perception might promote further understanding of a putative relationship between naturalness and self-relevance. We look forward to empirical contributions that increase our understanding of both self-voice processing and voice naturalness impressions.

Acknowledgments

We thank Manuel Pöhlmann and three anonymous reviewers for helpful suggestions on a previous version of this work. The authors gratefully acknowledge the award of funding through an EU-MSCA doctoral network "Voice Communication Sciences" (action 101168998).

Author Declarations

The authors have no conflicts to disclose**.**

Data Availability

There is no data linked to this manuscript.

References

DeBruine, L. M., Smith, F. G., Jones, B. C., Roberts, S. C., Petrie, M., & Spector, T. D. (2009). Kin recognition signals in adult faces. *Vision Res*, *49*(1), 38–43. https://doi.org/10.1016/j.visres.2008.09.025

Frühholz, S., & Schweinberger, S. R. (2021). Nonverbal auditory communication - Evidence for integrated neural systems for voice signal production and perception. *Progress in Neurobiology*, *199*, 101948. https://doi.org/10.1016/j.pneurobio.2020.101948

Hyppa-Martin, J., Lilley, J., Chen, M., Friese, J., Schmidt, C., & Bunnell, H. T. (2024). A large-scale comparison of two voice synthesis techniques on intelligibility, naturalness, preferences, and attitudes toward voices banked by individuals with amyotrophic lateral sclerosis. *Augmentative and Alternative Communication*, *40*(1), 31–45. https://doi.org/10.1080/07434618.2023.2262032

Kapolowicz, M. R., Guest, D. R., Montazeri, V., Baese-Berk, M. M., & Assmann, P. F. (2022). Effects of Spectral Envelope and Fundamental Frequency Shifts on the Perception of Foreign-Accented

Speech. *Language and Speech*, *65*(2), 418–443. https://doi.org/10.1177/00238309211029679

Kovács, G., Göschel, L., Pawlik, S. M., & Tramacere, A. (2025). The neural dynamics of current and past self-face perception: Challenging the privilege access hypothesis. *Cortex; a Journal Devoted to the Study of the Nervous System and Behavior*, *194*, 77–90. https://doi.org/10.1016/j.cortex.2025.11.006

Lavan, N., Irvine, M., Rosi, V., & McGettigan, C. (2025). Voice clones sound realistic but not (yet) hyperrealistic. *PLoS One*, *20*(9), e0332692. https://doi.org/10.1371/journal.pone.0332692

Lee, E.-J. (2010). The more humanlike, the better? How speech type and users' cognitive style affect social responses to computers. *Computers in Human Behavior*, *26*(4), 665–672. https://doi.org/10.1016/j.chb.2010.01.003

Maloney, L. T., & Dal Martello, M. F. (2006). Kin recognition and the perceived facial similarity of children. *Journal of Vision*, *6*(10), 1047–1056. https://doi.org/10.1167/6.10.4

Mateo, J. M. (2015). Perspectives: Hamilton's Legacy: Mechanisms of Kin Recognition in Humans. *Ethology*, *121*(5), 419–427. https://doi.org/10.1111/eth.12358

Maurer, D., & Landis, T. (1990). Role of bone conduction in the self-perception of speech. *Folia Phoniatrica*, *42*(5), 226–229. https://doi.org/10.1159/000266070

Nussbaum, C., Frühholz, S., & Schweinberger, S. R. (2025). Understanding voice naturalness. *Trends in Cognitive Sciences*, *29*(5), 467–480. https://doi.org/10.1016/j.tics.2025.01.010

Orepic, P., Kannape, O. A., Faivre, N., & Blanke, O. (2023). Bone conduction facilitates self-other voice discrimination. *Royal Society Open Science*, *10*(2), 221561. https://doi.org/10.1098/rsos.221561

Orepic, P., & Pinheiro, A. P. (2026). From Voice to Self: An Integrative Framework on Self-Voice Processing. *Perspect Psychol Sci*, 17456916261422585. https://doi.org/10.1177/17456916261422585

Pandey, A., Le Maguer, S., & Harte, N. (2025). What is Naturalness? In *13th edition of the Speech Synthesis Workshop* (pp. 215–221). ISCA. https://doi.org/10.21437/SSW.2025-33

Pinheiro, A. P. (2025). Beyond acoustics: Self-relevance as a key to voice naturalness (L). *The Journal of the Acoustical Society of America*, *158*(5), 4045–4047. https://doi.org/10.1121/10.0039927

Pinheiro, A. P., Sarzedas, J., Roberto, M. S., & Kotz, S. A. (2023). Attention and emotion shape self-voice prioritization in speech processing. *Cortex; a Journal Devoted to the Study of the Nervous System and Behavior*, *158*, 83–95. https://doi.org/10.1016/j.cortex.2022.10.006

Platek, S. M., & Kemp, S. M. (2009). Is family special to the brain? An event-related fMRI study of familiar, familial, and self-face recognition. *Neuropsychologia*, *47*(3), 849–858. https://doi.org/10.1016/j.neuropsychologia.2008.12.027

Roswandowitz, C., Kathiresan, T., Pellegrino, E., Dellwo, V., & Frühholz, S. (2024). Cortical-striatal brain network distinguishes deepfake from real speaker identity. *Communications Biology*, *7*(1), 711. https://doi.org/10.1038/s42003-024-06372-6

Sui, J., & Humphreys, G. W. (2017). The ubiquitous self: What the properties of self-bias tell us about the self. *Ann N Y Acad Sci*, *1396*(1), 222–235. https://doi.org/10.1111/nyas.13197

Tsuchiya, A., & Schweinberger, S. R. (2022). Erkennen von Verwandtschaft zu sich Selbst und bei Anderen [Recognition of Kinship to Self and in Others]: Part 2, Issue 1/2022. *In-Mind Magazine*, *1*(2). https://de.in-mind.org/article/erkennen-von-verwandtschaft-zu-sich-selbst-und-bei-anderen

Yamagishi, J., Veaux, C., King, S., & Renals, S. (2012). Speech synthesis technologies for individuals with vocal disabilities: Voice banking and reconstruction. *Acoustical Science and Technology*, *33*(1), 1–5. https://doi.org/10.1250/ast.33.1